

HDOC DAY 21_PROGRAMMING IN C

Problem

Write a menu-driven C program that accepts two positive integers, m and n , where $m \leq n$. Display the following menu and accept the user's choice. Use a switch statement to perform the selected operation:

- 1. Print the first 10 multiples of each integer from m to n .*
- 2. Print the addition table of each integer from m to n , using addends from 1 to 10.*
- 3. Print the subtraction table of each integer from m to n , using subtrahends from 1 to 10.*
- 4. Print the results obtained by dividing each integer from m to n by the integers from 1 to 10. Display the results rounded to two decimal places.*
- 5. Exit the program.*

Display an appropriate error message if the user enters an invalid menu choice. Continue displaying the menu and performing the selected operation until the user chooses the Exit option.

Algorithm

- 1. Start**
2. Initialize a variable $i = 0$.
3. Repeat the following steps while $i \neq 0$:
 1. Input two positive integers n and m , where n is larger than m .
 2. Display the menu with five choices:
 - Choice 1: Find the first 10 multiples of each integer from m to n .
 - Choice 2: Print the addition table of each integer from m to n , using addends from 1 to 10.
 - Choice 3: Print the subtraction table of each integer from m to n , using subtrahends from 1 to 10.
 - Choice 4: Divide each integer from m to n by integers from 1 to 10.
 - Choice 5: Exit the program.
 3. Input the user's choice ch .

4. Use a **switch statement** based on ch:
 - **Case 1: Multiplication**
 1. For each integer i from m to n, print its first 10 multiples.
 2. For j = 1 to 10, calculate $i \times j$.
 3. Display each result.
 - **Case 2: Addition**
 1. For each integer i from m to n, print its addition table.
 2. For j = 1 to 10, calculate $i + j$.
 3. Display each result.
 - **Case 3: Subtraction**
 1. For each integer i from m to n, print its subtraction table.
 2. For j = 1 to 10, calculate $i - j$.
 3. Display each result.
 - **Case 4: Division**
 1. For each integer i from m to n, print its division results.
 2. For j = 1 to 10, calculate i / j .
 3. Display each result as a floating-point value.
 - **Case 5: Exit**
 1. Terminate the program.
 - **Default**
 1. Display "Invalid choice".
 2. Return to the menu.
4. Continue displaying the menu until the user selects **Choice 5**.
5. **Stop**.

C PROGRAM CODE

```
#include <stdio.h>
```

```
int main()
```

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```
{int i=0;
while(i==0)
{
    int n,m,ch;

    printf("enter two positive numbers in order:1st larger and second smaller\n");
    scanf("%d%d",&n,&m);

    printf("choice 1 is to find first 10 multiples of each number from m to n\n");

    printf("choice 2 is to Print the addition table of each integer from m to n, using
addends from 1 to 10.\n");

    printf("choice 3 is toPrint the subtraction table of each integer from m to n, using
subtrahends from 1 to 10.\n");

    printf("choice 4 is to Print the results obtained by dividing each integer from m to n by
the integers from 1 to 10.\n");

    printf ("choice 5 is to exit program\n");

    printf("enter your choice");

    scanf("%d",&ch);

    switch (ch)
    {
        case 1:
        {
            for(int i=m;i<=n;i++)
            {printf("multiples of %d are\n",i);
                for(int j=1;j<=10;j++)
                {
                    printf("%d\t",(i*j));

                }
            }
            printf("\n");
        }
        break;
```

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```
}  
case 2:  
{  
    for(int i=m;i<=n;i++)  
    { printf("\nAddition table for %d:\n", i);  
        for(int j=1;j<=10;j++)  
        {  
            printf("%d+%d=%d\n",i,j,(i+j));  
        }  
    }  
    break;  
}  
case 3:  
{  
    for(int i=m;i<=n;i++)  
    { printf("subtraction table for %d:\n", i);  
        for(int j=1;j<=10;j++)  
        {  
            printf("%d-%d=%d\n",i,j,(i-j));  
        }  
    }  
    break;  
}  
case 4:  
{  
    for(int i=m;i<=n;i++)  
    { printf("division products for %d:\n", i);  
        for(int j=1;j<=10;j++)
```

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```
        {
            float div=i/j;
            printf("%d/%d=%2f\n",i,j,div);
        }
    }
    break;
}
case 5:
{
    return 0;
}
default:
{
    printf("invalid choice\n");
    break;
}
}
}
return 0;
}
```